

CHALLA VENKATA SAI GIRISH

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PROFESSIONAL SUMMARY

Final-year Computer Science student with 5 deployed production systems across backend engineering, NLP, computer vision, and combinatorial optimization. Proficient in Python, FastAPI, Flask, PostgreSQL, and Docker. Built and deployed systems with CI/CD pipelines, live Render deployments, and CI green checkmarks across all projects.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML

Backend: FastAPI, Flask, REST APIs, JWT Auth, RBAC, SocketIO

Database: PostgreSQL (Star Schema), MySQL, MongoDB

ML/AI: YOLOv5, scikit-learn (TF-IDF, LinearSVC), OR-Tools (CP-SAT, LNS), OpenCV, LSTM, Random Forest

DevOps: Docker, GitHub Actions, Render, CI/CD

PROJECTS

GlobalScart 360 — Backend Commerce Engine [Live Shop](#) · [GitHub](#) · [Case Study](#)

Python · FastAPI · PostgreSQL · JWT · Docker · GitHub Actions · Render

- Production-style backend with multi-stage checkout lifecycle (ORDER_CREATED → PAYMENT_PENDING → SUCCESS/FAIL) using PostgreSQL transactions for atomic consistency
- Implemented JWT-secured RBAC, row-level locking for inventory, and idempotency table for payment webhook retries
- Built near real-time analytics pipeline with star-schema, incremental upserts, and request-ID tracing middleware
- Fully containerized with Docker Compose; CI/CD via GitHub Actions with green checkmark on all pushes

University Timetabling Solver — CP-SAT + LNS Optimizer · [Dashboard](#)

Python · OR-Tools (CP-SAT) · FastAPI · Large Neighborhood Search · ITC-2019

- Hybrid CP-SAT + LNS optimizer that detects infeasible university timetables and auto-generates feasible schedules
- Benchmarked against ITC-2019 standard instances; achieves Quality Score 100 on test cases with 150 classes and 7 rooms
- Deployed FastAPI backend on Render with interactive GitHub Pages dashboard for live visualization

Hate Speech Detection — NLP Classifier + AI Chatroom [Live Demo](#) · [GitHub](#)

Python · Flask · TF-IDF · LinearSVC · SocketIO · scikit-learn · NLTK

- TF-IDF + LinearSVC text classifier achieving 78.5% accuracy (Macro F1: 0.78) on 6,424 test samples
- YouTube comment classifier detects toxic content with sentiment analysis and keyword highlighting
- Real-time AI polite chatroom that auto-detects hate speech via SocketIO and converts messages to polite alternatives

Power Theft Detection System — Smart Grid Monitoring [Live Dashboard](#) · [Case Study](#)

Python · Flask · Pandas · NumPy · Gunicorn · Render

- Prototype smart grid monitoring system with heuristic risk scoring (HIGH/MEDIUM/LOW) across 2,350 simulated customers
- Interactive dashboard with year-over-year comparison (2015–2025), real-time alerts, and power flow simulation
- Modular ML-ready architecture with offline scikit-learn models (LSTM, Random Forest) for future inference pipeline

Smart Marine AI — Marine Debris Detection [GitHub](#) · [Case Study](#)

Python · YOLOv5 · OpenCV · Streamlit · PyTorch

- YOLOv5m-based plastic waste detection system achieving 92% detection accuracy on marine debris dataset
- Real-time webcam detection, batch image processing, analytics dashboard, and GPS-based autonomous collection simulation

EDUCATION

B.Tech in Computer Science & Engineering — Anurag University, Hyderabad | CGPA: 7.1 | 2023 – 2026

Diploma in Electronics & Communication Engineering — Govt. Polytechnic Warangal | CGPA: 7.19 | 2020 – 2023